

FORTIGATE NSE4

Lab 4 — VLANs and Sub-interfaces

Router-on-a-Stick with Policy-Enforced Inter-VLAN Routing

1. Objective and Design

This lab implements router-on-a-stick on a FortiGate: two VLANs carried over a single physical link as 802.1Q sub-interfaces, with the firewall acting as the gateway for both.

The design requirement is that traffic between the two VLANs must pass through the firewall rather than being switched locally. The topology achieves this by construction rather than by configuration.

► 1.1 Why the Switch Is Deliberately Layer 2 Only

The switch carries no SVIs and has routing disabled, so the only Layer 3 interface for either subnet exists on the FortiGate. Inter-VLAN traffic has nowhere to go except up the trunk and through the policy engine.

Important: the alternative fails silently. If SVIs were configured on the switch, it would route between the VLANs locally and the FortiGate would never see the traffic — while every host would still report full connectivity. The security control would be absent and nothing would indicate it.

2. Lab Topology

A single FortiGate carries both VLANs over one physical link to a Layer 2 switch. Each VLAN terminates on its own 802.1Q sub-interface, and no Layer 3 interface for either subnet exists anywhere except on the firewall.

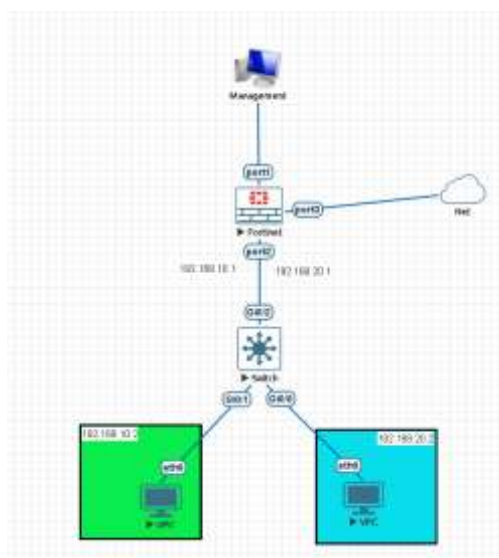


Figure 1 — Lab topology: both VLANs reach the FortiGate over a single trunk on port2

Because both subnets are directly connected on the FortiGate, no routing protocol is required — the firewall already holds a route to each. Separation between them is enforced entirely by 802.1Q tagging on the trunk and by firewall policy above it.

► 2.1 Addressing Plan

VLAN	Name	Subnet	Gateway	Host
10	Green	192.168.10.0/24	192.168.10.1	192.168.10.2
20	Blue	192.168.20.0/24	192.168.20.1	192.168.20.2
999	Native-Unused	—	—	Trunk native VLAN only

Port	Role
port1	Management access
port2	Trunk to the switch, carrying both VLANs as tagged sub-interfaces
port3	Upstream toward the external network

3. Switch Configuration

The switch is configured purely as a Layer 2 device: two access ports and one trunk.

```

vlan 10
  name Green
vlan 20
  name Blue
vlan 999
  name NATIVE-UNUSED
!
interface GigabitEthernet0/1
  switchport mode access
  switchport access vlan 10
  no shutdown
!
interface GigabitEthernet0/0
  switchport mode access
  switchport access vlan 20
  no shutdown
!
interface GigabitEthernet0/2
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport trunk allowed vlan 10,20
  switchport trunk native vlan 999
  no shutdown

```

► 3.1 Allowed VLAN List

A trunk permits all VLANs by default. Restricting it to 10 and 20 means a VLAN created later is not carried across the trunk unless it is added explicitly — the trunk stops being an implicit path for anything that happens to exist on the switch.

► 3.2 Unused Native VLAN

Frames in the native VLAN cross a trunk untagged. FortiGate sub-interfaces match on VLAN tag, so if VLAN 10 or 20 were left as native, that VLAN's frames would arrive without a tag and be discarded with no error.

4. FortiGate Sub-interfaces

Each VLAN is created as a sub-interface with `port2` as its parent. The VLAN ID must match the tag applied by the switch.

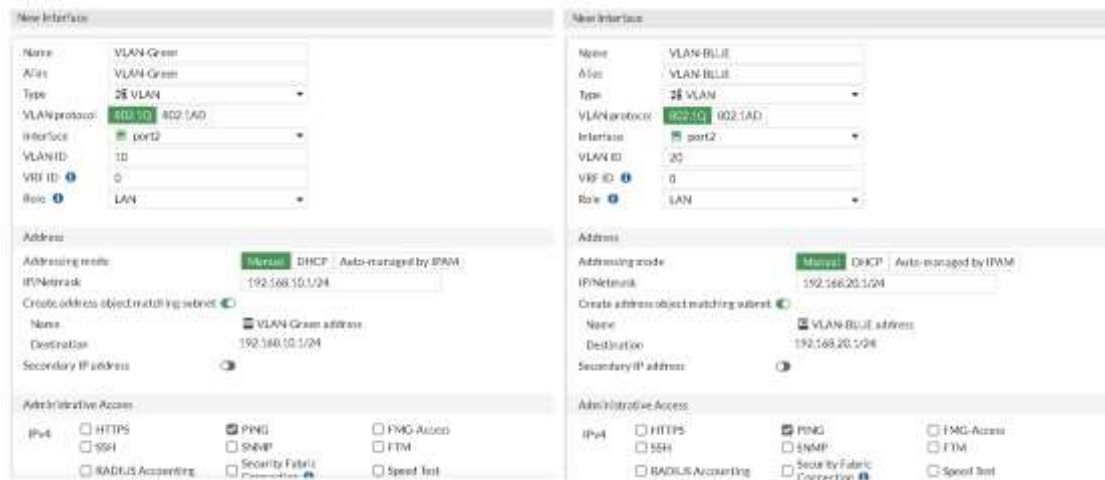


Figure 2 — VLAN-Green (VLAN 10) and VLAN-BLUE (VLAN 20), both parented to port2

Field	VLAN-Green	VLAN-BLUE
Type	VLAN	VLAN
VLAN protocol	802.1Q	802.1Q
Interface (parent)	port2	port2
VLAN ID	10	20
IP / Netmask	192.168.10.1/24	192.168.20.1/24
Role	LAN	LAN
Administrative Access	PING only	PING only

Administrative access is limited to PING on both. This is enough to verify the gateway is reachable without exposing HTTPS or SSH management to a user VLAN — an interface facing end hosts should not offer administrative protocols unless there is a specific reason.

The **Create address object matching subnet** option was enabled on both, which generates a firewall address object for each subnet automatically. Those objects are used directly in the policies that follow.

The image shows a screenshot of the FortiGate interface table. It lists 'port2' as the physical interface with IP 0.0.0.0/0.0.0.0. Below it, two sub-interfaces are listed: 'VLAN-BLUE (VLAN-BLUE)' with IP 192.168.20.1/255.255.255.0 and 'VLAN-Green (VLAN-Green)' with IP 192.168.10.1/255.255.255.0. Both are marked as 'PING'.

port2	Physical Interface	0.0.0.0/0.0.0.0	
VLAN-BLUE (VLAN-BLUE)	VLAN	192.168.20.1/255.255.255.0	PING
VLAN-Green (VLAN-Green)	VLAN	192.168.10.1/255.255.255.0	PING

Figure 3 — Both sub-interfaces listed under port2 in the interface table

5. Baseline Verification

Before adding any policy, each host was configured and tested against its own gateway.

```
VPCS> ip 192.168.10.2 255.255.255.0 192.168.10.1
Checking for duplicate address...
VPCS : 192.168.10.2 255.255.255.0 gateway 192.168.10.1

VPCS> ping 192.168.10.1

84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=3.269 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=1.691 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=3.848 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=5.044 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=4.616 ms
```

Figure 4 — VPC in VLAN 10 reaching its gateway

```
VPCS> ip 192.168.20.2 255.255.255.0 192.168.20.1
Checking for duplicate address...
VPCS : 192.168.20.2 255.255.255.0 gateway 192.168.20.1

VPCS> ping 192.168.20.1

84 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=3.821 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=1.994 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=7.602 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=255 time=4.763 ms
84 bytes from 192.168.20.1 icmp_seq=5 ttl=255 time=3.453 ms
```

Figure 5 — VPC in VLAN 20 reaching its gateway

Both succeed, which confirms three things at once: the access ports are in the correct VLANs, the switch is tagging correctly on the trunk, and the FortiGate sub-interfaces are matching those tags. A failure here would point at VLAN configuration rather than at policy.

► 5.1 Confirming the Default Deny

Cross-VLAN traffic was then tested — before any policy exists.

```
VPCS> ping 192.168.10.1

192.168.10.1 icmp_seq=1 timeout
192.168.10.1 icmp_seq=2 timeout
192.168.10.1 icmp_seq=3 timeout
```

```
VPCS> ping 192.168.10.2

192.168.10.2 icmp_seq=1 timeout
192.168.10.2 icmp_seq=2 timeout
```

```
VPCS> ping 192.168.20.2

192.168.20.2 icmp_seq=1 timeout
192.168.20.2 icmp_seq=2 timeout
192.168.20.2 icmp_seq=3 timeout
```

Figure 6 — All cross-VLAN traffic times out: the far gateway, and each host from the other VLAN

Every cross-VLAN attempt fails. This is the implicit deny rule working as intended, and it is worth capturing deliberately rather than assuming.

Why this step matters: the baseline proves the requirement is met. If inter-VLAN traffic passed at this stage, it would mean the traffic was being routed somewhere other than the FortiGate — and the entire premise of the design would be broken. A test that succeeds later only demonstrates the security control is working if you have first established that it fails without one.

6. Inter-VLAN Policy

With the baseline established, a policy was created to permit ICMP from the Green VLAN to the Blue VLAN.

The screenshot shows the 'New Policy' configuration window in FortiGate. The policy name is 'VLAN-Green to VLAN-Blue'. The incoming interface is 'VLAN-Green (VLAN-Green)' and the outgoing interface is 'VLAN-BLUE (VLAN-BLUE)'. The source is 'VLAN-Green address' and the destination is 'VLAN-BLUE address'. The schedule is set to 'always' and the service is 'PING'. The action is 'ACCEPT'. The inspection mode is 'Flow-based'. The NAT is disabled. The protocol options are set to 'not default'.

Figure 7 — Policy permitting PING from VLAN-Green to VLAN-BLUE

Setting	Value	Reasoning
Incoming / Outgoing	VLAN-Green → VLAN-BLUE	Sub-interfaces are selected as interfaces in their own right
Source / Destination	VLAN-Green address / VLAN-BLUE address	The auto-generated subnet objects, rather than 'all'
Service	PING	Scoped to the protocol being tested rather than opened to ALL
Action	ACCEPT	—
NAT	Disabled	Deliberate — see 6.1

► 6.1 Why NAT Is Off for Inter-VLAN Traffic

Enabling NAT here would translate the source to the FortiGate's VLAN-BLUE address, so every Green host would appear to the Blue VLAN as 192.168.20.1. Hosts could not be distinguished from one another, and the forward traffic log would attribute every session to the firewall instead of to the host that originated it.

Rule: NAT belongs on the path to an external network where address translation is required. Between internal VLANs it destroys attribution for no benefit.

```

VPCS> ping 192.168.20.2

192.168.20.2 icmp_seq=1 timeout
192.168.20.2 icmp_seq=2 timeout
192.168.20.2 icmp_seq=3 timeout
192.168.20.2 icmp_seq=4 timeout
192.168.20.2 icmp_seq=5 timeout

VPCS> ping 192.168.20.2

84 bytes from 192.168.20.2 icmp_seq=1 ttl=63 time=19.714 ms
84 bytes from 192.168.20.2 icmp_seq=2 ttl=63 time=9.518 ms
84 bytes from 192.168.20.2 icmp_seq=3 ttl=63 time=4.374 ms
84 bytes from 192.168.20.2 icmp_seq=4 ttl=63 time=8.629 ms
84 bytes from 192.168.20.2 icmp_seq=5 ttl=63 time=7.977 ms

```

Figure 8 — The same ping before and after the policy: timeout, then replies

The result is unambiguous. The identical command fails before the policy and succeeds after it. The TTL is also worth noting — replies from a host across the VLAN boundary return with `ttl=63` rather than the `ttl=255` seen when pinging the local gateway. The decrement is the FortiGate routing the packet, which is direct evidence that the traffic was forwarded at Layer 3 rather than switched.

7. Policy Directionality

A ping was then attempted in the reverse direction, from the Blue VLAN toward the Green VLAN.

```

VPCS> ping 192.168.10.2

192.168.10.2 icmp_seq=1 timeout
192.168.10.2 icmp_seq=2 timeout
192.168.10.2 icmp_seq=3 timeout

```

Figure 9 — The reverse direction still fails — no policy permits it

It fails, because FortiOS policies are unidirectional. A policy from VLAN-Green to VLAN-BLUE permits sessions initiated from Green only.

8. External Access

The final stage routes VLAN traffic toward the external network through port3.

► 8.1 Static Route

Destination #	Gateway IP #	Interface #	Status #	Comments #
8.8.8.8/24	172.16.0.1	port3	Enabled	

Figure 10 — Static route toward the external destination via port3

The route directs the target network to the upstream next hop on port3. A production configuration would normally use a default route; scoping the route to a specific destination keeps the lab's forwarding behaviour narrow and explicit.

► 8.2 Address Object

Figure 11 — A subnet address object for the external destination

Defining a named object rather than using 'all' as the destination keeps the policy scoped to what is actually being permitted, and makes the intent readable in the policy list.

► 8.3 Outbound Policy

Figure 12 — Outbound policy from VLAN-BLUE to port3 with NAT enabled

Setting	Value	Reasoning
Incoming / Outgoing	VLAN-BLUE → port3	Internal VLAN toward the upstream interface
Source / Destination	VLAN-BLUE address / the external object	Both scoped to named objects
Service	PING	Limited to what is being tested
NAT	Enabled	Required — private VLAN addressing is not routable upstream
IP Pool Configuration	Use Outgoing Interface Address	Translates to port3's address; the standard choice

NAT is enabled here for the opposite reason it was disabled on the inter-VLAN policy: the internal addressing is private and cannot be routed beyond the FortiGate without translation.

9. Summary

Objective	Result
Two VLANs carried over a single trunk	Achieved — 802.1Q sub-interfaces on port2, both hosts reach their gateways
Inter-VLAN traffic forced through the firewall	Achieved — switch is Layer 2 only; all cross-VLAN traffic failed until a policy permitted it
Controlled inter-VLAN communication	Achieved — PING permitted from Green to Blue; reverse direction still denied
Outbound access to an external network	Configuration verified;